

Assignment- Day 2

Query Preprocessing:

Navigation

Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion re-ranking

Summarize & Dedup

AI summarization & deduplication

Prompt & Schema

Configure prompt templates & JSON schemas

Settings

Configure environment

RAG Pipeline

Home > Query Preprocessing

Query Preprocessing & Search

Transform your query through normalization, abbreviation expansion, and synonym generation before searching

Enter Your Query

password reset OTP verification

Quick Examples

UHD patient login issue OTP not working

TC_027 r

PREPROCESS QUERY

PIPELINE STEPS

ABBREVIATIONS

SYNONYM VARIATIONS

Query Transformation Pipeline

ORIGINAL QUERY

"password reset OTP verification"

NORMALIZATION

"password reset otp verification"

4 tokens

ABBREVIATION EXPANSION

1 expansions

"password reset one time password verification"

Expanded:

otp -> one time password

SYNONYM EXPANSION

5 variations

Multiple query variations generated for comprehensive search coverage

1. "password reset one time password verification"

2. "credential reset one time password verification"

3. "passphrase reset one time password verification"

+ 2 more variations

RAG Demo v1.2

Vector Search:

Navigation

Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion re-ranking

Summarize & Dedup

AI summarization & deduplication

Prompt & Schema

Configure prompt templates & JSON schemas

Settings

Configure environment

RAG Pipeline

Home > Vector Search

Query Search

Search through your test cases using semantic similarity with AI embeddings.

Search Query

Enter your search terms and find similar test cases

Share IP reports

Use descriptive terms related to test functionality

Advanced Metadata Filters

Module

Priority

Risk Level

Test Type

HIDE FILTERS

Results Limit

5 Results

SEARCH

Search Query: "Share IP reports"

Performance: (\$0.000001 cost) 5 tokens

5 results

Search Results

5 results ranked by semantic similarity

Test Case	Similarity	Description	Source
Reports ->View Reports->Summary Reports-> Joint Bill	87.8%	Verify user is able to view Joint Billing Summary(IP)	
IP Billing			
Reports ->View Reports->Deposit Report ->IP Deposit	87.8%	Verify user is able to view IP Deposit Report	
IP Billing			
Consultant Dashboard - IP Report	87.8%	Verify user able to view IP Report	
Doctors Wards			
Doctor Dashboard - IP Report	87.8%	Verify user able to view IP Report	
Doctors Wards			

Rows per page: 10 1-5 of 5

Detailed View

RAG Demo v1.2

Enterprise Edition

Vector Search with metadata filters:

Navigation

Convert to JSON

Upload and convert Excel files
(Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization & deduplication

Prompt & Schema

Configure prompt templates & JSON schemas

Settings

Configure environment

RAG Demo v1.2

Enterprise Edition

< RAG Pipeline

Home > Vector Search

Use descriptive terms related to test functionality

Advanced Metadata Filters

Active

Module

IP Billing

Priority

Risk Level

Test Type

Active Filters: Module: IP Billing

SEARCH

Search Query: "Share IP reports"

Performance: \$0.000001 cost 5 tokens

Applied Filters: module: IP Billing

Search Results

5 results ranked by semantic similarity

Test Case	Similarity	Description
Reports ->View Reports->Deposit Report ->IP Deposit	87.8%	Verify user is able to view IP Deposit Report
IP Billing		
Reports-Admin Reports-Others- Joint Billing Summary	87.2%	Verify User able to View Joint Billing Summary-PDF
IP Billing		
Reports ->View Reports->Summary Reports-> Joint R	87.1%	Verify user is able to view Joint Billing Summary
IP Billing		
Reports-Admin Reports-Others- OP services Against I	87.0%	Verify User able to view OP services Against IP
IP Billing		

BM25 Search:

Navigation

Convert to JSON

Upload and convert Excel files
(Test Cases or User Stories)

Embeddings & Store

Create and manage
embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization &
deduplication

Prompt & Schema

Configure prompt templates &
JSON schemas

Settings

Configure environment

RAG Demo v1.2

Enterprise Edition

< RAG Pipeline

Home > BM25 Search

Keywords

BM25 Keyword Search

Fast keyword-based search using BM25 algorithm. Best for exact matches, IDs, and specific terms.

Search Query

Login failed

SHOW FILTERS

Results L...

10

SEARCH

BM25 Search Results: Found 10 test cases in 343ms

Search Type: bm25

Search Results (10)

#1 TC_2616

Score: 9.87

Mastersetup->Map Login Counter

Module: AHC P1 Risk: Critical Automated

Description: To verify the user able to create Map Login Counter.

Steps: 1.Launch the application 2.Login to the application with valid credentials. 3.Click on Patient Service Tab and select AHC from drop down. 4.Navigate to Functions -> Mastersetup ->Map Login Counter. 5.Click on Map Login Counter. 6.Enter Login ID and select Counter. 7.Click on "ADD" button.

Expected Results: User should able to create Map Login Counter.

Created: 7/25/2026

#2 TC_2617

Score: 9.63

Mastersetup->Map Login Counter ->Delete

Module: AHC P1 Risk: Critical Automated

Description: To verify the user able to delete Map Login Counter.

Navigation

Convert to JSON

Upload and convert Excel files
(Test Cases or User Stories)

Embeddings & Store

Create and manage
embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization &
deduplication

Prompt & Schema

Configure prompt templates &
JSON schemas

Settings

Configure environment

RAG Demo v1.2

Enterprise Edition

< RAG Pipeline

Home > BM25 Search

Keywords

BM25 Keyword Search

Fast keyword-based search using BM25 algorithm. Best for exact matches, IDs, and specific terms.

Search Query

SHOW FILTERS

Results L...

SEARCH

BM25 Search Results: Found 10 test cases in 343ms

Search Type: bm25

Search Results (10)

#8 TC_3204

Score: 4.99

Masters ->Service Definitions ->Logic Bases Definition

Module: IP Billing P1 Risk: High Automated

Description: Verify user able to set logic based Definition

Steps: 1. Launch the Application. 2.Login to the application with valid credentials 3.Click on Common Support Services Tab and select Billing from the drop down. 4.Navigate to Functions ->masters ->Service Definitions ->Logic Bases Definition. 5.Select service item and customer name and click on search. 6.Under logic tab, enter from search, to search, effective from date, effective to date, select unit of measurement, amount, click on add button. select slot type radio button and click on save button.

Expected Results: User can able to see the PopUp Message like Role ResourceItem(s) updated successfully.

Created: 7/25/2026

#9 TC_4582

Score: 4.03

Wards Login

Module: Doctors-Wards P1 Risk: Critical Automated

Description: Verify User Able to view Ward Details

Steps: 1.Launch the application 2.Login to the application with valid credentials. 3.Click on Patient Service Tab and select Doctors and wards from the drop down. 4.Navigate to Functions->Transactions->Wards Login 5.Select Ward

Expected Results: The Selected Ward Bed Details, New Admissions and Marked For Discharge details should be displayed

Created: 7/25/2026

#10 TC_2399

Score: 3.78

Login Details Reports

Module: Admin P1 Risk: Critical Automated

Description: To verify the user able to Delete the role.

Steps: 1.Launch the application 2.Login to the application with valid credentials. 3.Click on External Service Tab and select Admin from the drop down. 4.Click on Navigation Menu -> Sessions ->Login Details Reports. 5.Click on Login Details Reports.

Expected Results: User can able to see the PopUp Message like Role ResourceItem(s) updated successfully.

Created: 7/25/2026

Hybrid Search:

Navigation

Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization & deduplication

Prompt & Schema

Configure prompt templates & JSON schemas

Settings

Configure environment

RAG Demo v1.2

Enterprise Edition

<

RAG Pipeline

Home

>

Hybrid Search

Hybrid Search (BM25 + Vector)

Combines BM25 keyword matching with semantic vector search for best results. Adjust weights to control the balance.

Search Query

Share IP reports

Results Limit

10

SEARCH

Search Weights

BM25 Weight: 30%

Vector Weight: 70%

0% 50% 100%0% 50% 100%

BALANCED (50/50)

KEYWORD HEAVY (70/30)

SEMANTIC HEAVY (30/70)

SHOW FILTERS

Hybrid Search Results: Found 10 test cases in 774ms

Stats: 10 in both indexes,0 BM25 only,0 vector only

Timing: BM25: 94ms, Vector: 672ms

Cost: \$0.000001 (\$ tokens)

Search Results (10)

TC_3542

Hybrid: 91.5%

Found in: both

Reports ->View Reports->Deposit Report ->IP Deposit

Module: IP Billing

P1

Risk: High

Automated

Description: Verify user is able to view IP Deposit Report

Steps: 1. Launch the Application. 2.Login to the application with valid credentials 3.Click on Common Support Services Tab and select Billing from the drop down. 4.Navigate to Functions ->Reports ->View Reports->Deposit R 5.Enter ip no, counter no, shift number, from date, to date, select grouping, enter report heading, report format and click on view report button.

Expected Results: User should be able to view IP Deposit report.

Reranking:

Navigation

Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization & deduplication

Prompt & Schema

Configure prompt templates & JSON schemas

Settings

Configure environment

RAG Demo v1.2

Enterprise Edition

<

RAG Pipeline

Home

>

Score Fusion

BM25+Vector fusion reranking

Groq AI Reranking Search

Combines BM25 keyword search and Vector semantic search, then uses Groq AI for intelligent semantic reranking and relevance optimization.

Search Query

Share IP reports

Top K Can...

10

Final Limit

10

SEARCH

FILTERS

Groq AI Reranking Results: Found 10 test cases ranked by Groq AI

Timing: Total: 3977ms

Method: BM25 + Vector -> Groq AI Semantic Reranking

AFTER RERANKING (FINAL)

BEFORE RERANKING (ORIGINAL)

COMPARISON TABLE

After Groq AI Reranking (10 results)

TC_3542

Rerank Score: 0.0000

Reports ->View Reports->Deposit Report ->IP Deposit

Module: IP Billing

P1

Risk: High

Description: Verify user is able to view IP Deposit Report

Ranking Scores:

BM25 Score: 13.6484

Groq Rerank: 0.5000

TC_704

Rerank Score: 0.0000

Reports ->View Reports->Deposit Report ->IP Deposit

Summarization:

Navigation

Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization & deduplication

Prompt & Schema

Configure prompt templates & JSON schemas

Settings

Configure environment

RAG Demo v1.2

Enterprise Edition

RAG Pipeline

Home > Summarize & Dedup

AI summarization & deduplication

Search Summarization & Deduplication

Search, remove duplicates, and generate AI-powered summaries of test cases

Search Query

password reset OTP

Search Type

Vector

Limit

20

Quick Examples:

patient login authentication

merge UHID records

appointment booking

password reset OTP

SEARCH

DEDUPPLICATE

SUMMARIZE WITH AI

COPY ALL RESULTS

Dedup Threshold

0.85

Summary Type

Concise

Show Duplicates

SEARCH RESULTS

DEDUPPLICATED

AI SUMMARY

Deduplication Statistics

Original Count

20

After Deduplication

18

Duplicates Removed

2

Reduction

10.0%

Unique Results (18)

COPY ALL UNIQUE

TC_1742

✓ Unique

Score: 0.8664

OT

To verify the user able to Add the role to OT.

Admin

P4 Priority

Functional

TC_6134

✓ Unique

Score: 0.8643

Active Directory -> Search In AD->Reset password

Verify user should able to Reset password of Active Directory details

Human Resources (HR)

P3 Priority

Functional

TC_2421

✓ Unique

Score: 0.8615

Support ->OTCancellation -> OT Checkout

To verify the user able OT Checkout

Admin

P4 Priority

Functional

Navigation

Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization & deduplication

Prompt & Schema

Configure prompt templates & JSON schemas

Settings

Configure environment

RAG Demo v1.2

Enterprise Edition

RAG Pipeline

Home > Summarize & Dedup

AI summarization & deduplication

Search Summarization & Deduplication

Search, remove duplicates, and generate AI-powered summaries of test cases

Search Query

password reset OTP

Search Type

Vector

Limit

20

Quick Examples:

patient login authentication

merge UHID records

appointment booking

password reset OTP

SEARCH

DEDUPPLICATE

SUMMARIZE WITH AI

COPY ALL RESULTS

Dedup Threshold

0.85

Summary Type

Concise

Show Duplicates

SEARCH RESULTS

DEDUPPLICATED

AI SUMMARY

AI-Generated Summary

Search Results Overview -- "test cases"

[# | Test-Case ID | Title (Key Action) | Module | Core Purpose | Relevance |]

[1] | **TC_1742** | OT - Add Role | Admin | Verify a user can add a role in OT | 86.6 % |

[2] | **TC_6134** | AD - Search->Reset Password | Human Resources (HR) | Verify password reset for AD entries | 86.4 % |

[3] | **TC_2421** | OT Cancellation -> OT Checkout | Admin | Verify OT checkout functionality | 86.1 % |

[4] | **TC_2420** | OT Cancellation -> OT Consumption Cancellation | Admin | Verify OT consumption cancellation | 85.9 % |

[5] | **TC_498** | Billing Execution -> OT Clearance (Yes) | OP Billing | Verify OT clearance when "Yes" selected | 85.8 % |

[6] | **TC_066** | Generate OPD | Registration | Verify OPD generation | 85.7 % |

[7] | **TC_2604** | OT Graph | OT (Operation Theatre) & Cath | Verify OT graph/chart feature | 85.7 % |

[8] | **TC_505** | Billing Execution -> Collection (EFT) | OP Billing | Verify collection submission with EFT payment mode | 85.5 % |

Generation:

Navigation

Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization & deduplication

Prompt & Schema

Configure prompt templates & JSON schemas

Settings

Configure environment

RAG Demo v1.2

<

RAG Pipeline

Configure prompt templates & JSON schemas

Home > Prompt & Schema

Prompt Template & JSON Schema Manager

Configure and test AI prompt templates and JSON schemas for healthcare test cases

ICEPOT Framework Enabled: This manager now uses the ICEPOT methodology for structured prompt engineering: Instruction, Context, Examples, Persona, Output, Tone

JSON SCHEMA

PROMPT TEMPLATES

TEST & PREVIEW

METRICS EVALUATION

Test Prompt Template

Test your prompt template with sample data to see how the AI responds

Test Query / User Story

Module: Patient Communication & Diagnostics

User Story ID: HC-125

User Story Title: Share Diagnostic Reports with Patients via WhatsApp

Sample Test Cases (JSON)

```
[
  {
    "testCaseId": "TC_305",
    "module": "Laboratory",
    "testCaseTitle": "Lab Report Email Notification - Validated Results",
    "testCaseDescription": "Verify that the system sends email notification to patients when lab test results are validated and marked as complete by the technician.",
    "testSteps": [
      "1. Technician logs into Laboratory module",
      "2. Select pending test for Patient UHID: 123456-7890",
      "3. Enter test results in result_values field",
      "4. Click 'Validate Results' button",
      "5. System marks test status as 'VALIDATED'",
      "6. Verify email sent to patient registered email",
      "7. Verify notification logged in patient_notifications table"
    ]
  }
]
```

COMPLETE RAG PIPELINE

SAVE CONFIGURATION

Navigation

Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization & deduplication

Prompt & Schema

Configure prompt templates & JSON schemas

Settings

Configure environment

RAG Demo v1.2

Enterprise Edition

<

RAG Pipeline

Configure prompt templates & JSON schemas

Home > Prompt & Schema

Complete RAG Pipeline Results (10-Step Process)

1. User Input → 2. Preprocessing → 3. Hybrid Search → 3.5. User Story Search → 4. RRF Rerank → 5. Dedup → 6. Summarize → 7. Prompt → 8. Generate → 9. Validate → 10. HTML 50 hybrid search results

RRF Re-Ranking Applied 2 duplicates removed Top 8 results Validation Passed

10-Step Pipeline Completed

User Story Input Query Preprocessing (Normalize → Abbreviations → Synonyms) Hybrid Search - Test Cases (BM25 + Vector, weighted fusion) Hybrid Search - User Stories (BM25 + Vector) RRF Re-Ranking (Cross-encoder, top 10 selected) Deduplication (Cosine > 0.95) Summarization (TestLeaf API) Prompt Template + Context (ICEPOT framework) LLM Generation (TestLeaf API) JSON Validation (AJV) HTML Conversion (UI rendering)

Query Preprocessing: Applied transformations including synonym expansion, abbreviation handling, and healthcare domain optimization.

REFERENCE TEST CASES GENERATED TEST CASES AFFECTED USER STORIES (10)

BACK NEXT: GENERATED REGENERATE

Reference Test Cases (Retrieved from Database)

These are the top 8 most relevant test cases retrieved from the database that were used as context for generation.

Test Case ID	Title	Module	Priority	Preconditions	Test Steps	Expected Results	Score
TC_2736	AHCSummary->AHCSummary ->Diagnostic & Imaging report	AHC	P3	N/A	1. Launch the application 2. Login to the application with valid credentials. 3. Click on Patient Service Tab and select AHC from drop down. 4. Navigate to Functions-> AHCSummary ->AHCSummary. 5. Select date or Enter UHID/AHC No and click on 'Go' button. 6. Click on AHC Number. 7. Click on Diagnostic & Imaging link 8. Observe the results 9. Click on Download or print icon	Diagnostic & Imaging details should be displayed	N/A

localhost:3000

Chat

Navigation

Convert to JSON

Upload and convert Excel files (Test Cases or User Stories)

Embeddings & Store

Create and manage embeddings

Query Preprocessing

Transform & expand queries

Vector Search

Semantic vector search

BM25 Search

Keyword-based search

Hybrid Search

Combined BM25 + Vector

Score Fusion

BM25+Vector fusion reranking

Summarize & Dedup

AI summarization & deduplication

Prompt & Schema

Configure prompt templates & JSON schemas

Settings

Configure environment

RAG Demo v1.2

Enterprise Edition

RAG Pipeline

Home > Prompt & Schema

Configure prompt templates & JSON schemas

TC_2736

AHCSummary -> AHCSummary -> Diagnostic & Imaging report

AHC

P3

N/A

1. Launch the application

2. Login to the application with valid credentials.

3. Click on Patient Service Tab and select AHC from drop down.

4. Navigate to Functions -> AHCSummary -> AHCSummary.

5. Select date or Enter UHID/AHC No and click on 'Go' button.

6. Click on AHC Number.

7. Click on Diagnostic & Imaging link

8. Observe the results

9. Click on Download or print icon

Diagnostic & Imaging details should be displayed

N/A

TC_3672

Reports-Admin Reports-Others- Patient Form 60 Report

IP Billing

P1

N/A

1. Launch the application

2. Login to the application with valid credentials.

3. Click on Common Support Services Tab and select Billing from the drop down.

4. Navigate to Functions -> Reports -> Admin Reports

5. Navigate to Others Tab and Click on Patient Form 60 Report

6. Enter Search Criteria and Click on View

Patient Form 60 Report Should be displayed

N/A

Transfer Request within hospital

1. Launch the application

2. Login to the application with valid credentials.

3. Click on Laboratory tab from Patient Service Tab.

4. Click Pre Analytical Phase-> Transfer Test Within Health Care

User should be able to create

Total Test Cases: 3

Modules Represented: 7

Priority Distribution: P1: 1 | P2: 5

Avg Relevance Score: 0.000

Cost Breakdown: Generation: 8090 tokens (\$0.000641) | RAG Summary: 2850 tokens (\$0.000000) | Total: \$0.000641